Shenghan **Gao**

PHOTONIC INTEGRATED CIRCUITS · PHOTONICS HARDWARE · VALIDATION & TEST

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Summary

Ph.D. candidate in Electrical and Computer Engineering specializing in integrated photonics, photonic hardware validation, and coherent electro-optic systems. End-to-end experience across silicon photonics and TFLN/LNOI PIC design, passive and active photonic devices, cleanroom fabrication, fiber-to-chip integration, RF/EO characterization, DSP-based signal analysis, and Python/PyVISA/SCPI automated measurement. Seeking PIC design, silicon photonics, photonics validation/test, electro-optic device, and optical hardware R&D roles.

Core Skills

PIC design	Silicon photonics and TFLN/LNOI PIC design; waveguides, directional couplers, MMI/splitters, grating/edge couplers, interferometers, electro-optic modulators, thermal phase shifters
Simulation/layout	GDSFactory, Cadence Virtuoso, KLayout; Lumerical, RSoft, COMSOL, Zemax/CodeV; optical mode analysis, device optimization, PIC layout preparation
Fab/integration	Cleanroom fabrication, spin coating, photoresist processing, electron-beam lithography, etching, SEM inspection; fiber-array coupling, fiber-to-chip coupling, photonic packaging, optical alignment
Validation	Device characterization, high-speed EO/RF characterization, VNA, RF detection, high-speed/balanced photodetection, EOMs, OSAs, SVAs, spectrum analyzers, oscilloscopes, tunable lasers
Automation/DSP	Python measurement automation, PyVISA/SCPI, NI MAX, automated testing and data acquisition, parameter sweeps, data logging, FFT/frequency-domain analysis, DSP equalization, MATLAB
Coherent optics	Optical injection locking, semiconductor-laser rate equations, coherent interference, carrier recovery, AM/PM modulation transfer, PID phase stabilization

Research and Education Experience

University of Pittsburgh — *Pittsburgh, PA*

PH.D. RESEARCH ASSISTANT | ADVISOR: NATHAN YOUNGBLOOD — *Spring 2020 - present*

- Developed an optical-injection-locking framework for distributed coherent photonic computing, combining semiconductor-laser rate-equation modeling, injection-ratio/detuning sweeps, coherent carrier recovery, and balanced coherent detection.
- Designed chip-scale photonic integrated circuits to translate fiber injection-locking systems onto integrated platforms, including waveguides, directional couplers, MMI/splitters, grating couplers, edge-coupling interfaces, and interferometric paths.
- Simulated silicon photonics and TFLN/LNOI passive and active photonic devices using GDSFactory, Cadence Virtuoso, KLayout, Lumerical, RSoft, and COMSOL workflows for layout, optical mode analysis, electro-optic modulation, and device optimization.
- Performed and supported photonic cleanroom fabrication and process evaluation, including spin coating/photoresist processing, e-beam lithography, etching, SEM inspection, and characterization of passive and active photonic structures.
- Integrated and aligned photonic chips using fiber arrays, grating-coupler alignment, edge coupling, optical packaging, VOAs, polarization controllers, and tunable-laser setups to connect fabricated devices to validation testbeds.
- Built high-speed EO/RF validation workflows using EOMs, high-speed photodetectors, balanced photodetection, OSAs, SVAs, spectrum analyzers, VNA-based characterization, RF detection, lock-in detection, PID phase feedback, and fiber-stretcher control.
- Automated photonic hardware validation with Python, PyVISA/SCPI, and NI MAX for wavelength, power, injection-ratio, and frequency sweeps; logged data and generated FFT/DSP analysis, plots, and repeatable reports.

University of Rochester — *Rochester, NY*

RESEARCH ASSISTANT | ADVISOR: XI-CHENG ZHANG — *Sep 2017 - Aug 2019*

- Generated THz signals from liquid plasmas and quantified how temperature, polarity, and viscosity affected emission strength, bandwidth, repeatability, and spectral content.
- Processed time- and frequency-domain experimental data using SNR, sensitivity, regression, and repeatability analysis; prototyped digital cloaking with lenslet arrays and MATLAB pixel remapping.

University of Pittsburgh — *Pittsburgh, PA*

PH.D. IN ELECTRICAL AND COMPUTER ENGINEERING — *Dec. 2026 expected*

Georgia Institute of Technology — *Atlanta, GA*

M.S. IN COMPUTER SCIENCE, MACHINE LEARNING TRACK — *May 2027 expected*

University of Rochester — *Rochester, NY*

M.S. IN OPTICS (THESIS TRACK) — *Aug. 2019*

B.S. IN OPTICAL ENGINEERING — *May 2017*

B.S. IN APPLIED MATHEMATICS — *May 2017*

Industry Experience

Lumentum Operations LLC *San Jose, CA*

Research Intern | Mentor: Joe Wen *May 2023 - Aug 2023*

- Developed MATLAB/Python FIR-filter equalization workflows for communication signals distorted by intersymbol interference, connecting DSP filter design to recovered waveform quality.
- Collected measured signal data and applied FFT/frequency-domain analysis to compare filter responses, tune coefficients, and validate demodulated-signal recovery across test cases.

Sunny Optics *Ningbo, China*

Research Intern | Mentor: Chao Xu *May 2018 - Aug 2018*

- Designed and optimized automotive LiDAR and mobile-camera lens systems in CodeV/Zemax against optical performance, tolerance, and packaging requirements.
- Collaborated with opto-mechanical engineers during lab integration, using alignment and test feedback to refine optical system functionality.

Selected Publications and Talks

MOSAIC: A Multicore Optical Architecture for Scaling Up AI Acceleration

L. Liu, R. Zhu, **S. Gao**, X. Xin, Y. Zhang, N. Youngblood, J. Yang, submitted to HPCA 2027, under review, 2026

Distributed Coherent Optical Computing via Injection-Locked Photonic Networks

Gao, S., K. Lüdge, F. Da Ros, N. Youngblood, submitted to APL, under review, 2026

Distributed Coherent Photonic Computing via Injection Locking

Gao, S., SPIE Optics + Photonics (oral presentation), 2026

Distributed Coherent Optical Computing via Injection-Locked Photonic Networks

Gao, S., Invited PhD Candidate Talk, Pittsburgh Quantum Institute, Pittsburgh, 2025

Strain-free FBG sensors for accurate temperature measurements from cryogenic temperatures to 300 K

Gao, S., P. R. Swinehart, S. Zhong, J. Wuenschell, N. Lalam, M. Buric, K. P. Chen, SPIE Defense + Commercial Sensing (oral presentation), 2023

Preference of subpicosecond laser pulses for terahertz wave generation from liquids

Q. Jin, Y. E., **S. Gao**, X.-C. Zhang, *Advanced Photonics* 2(1), 015001, 2020

Investigation of liquid properties on emitting terahertz wave under ultrashort optical excitation

S. Gao, Q. Jin, Y. E., X.-C. Zhang, SPIE, 2019